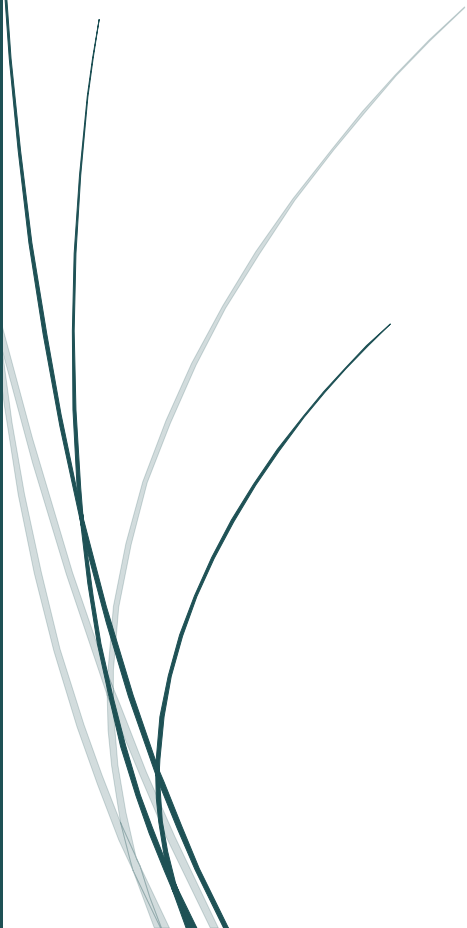




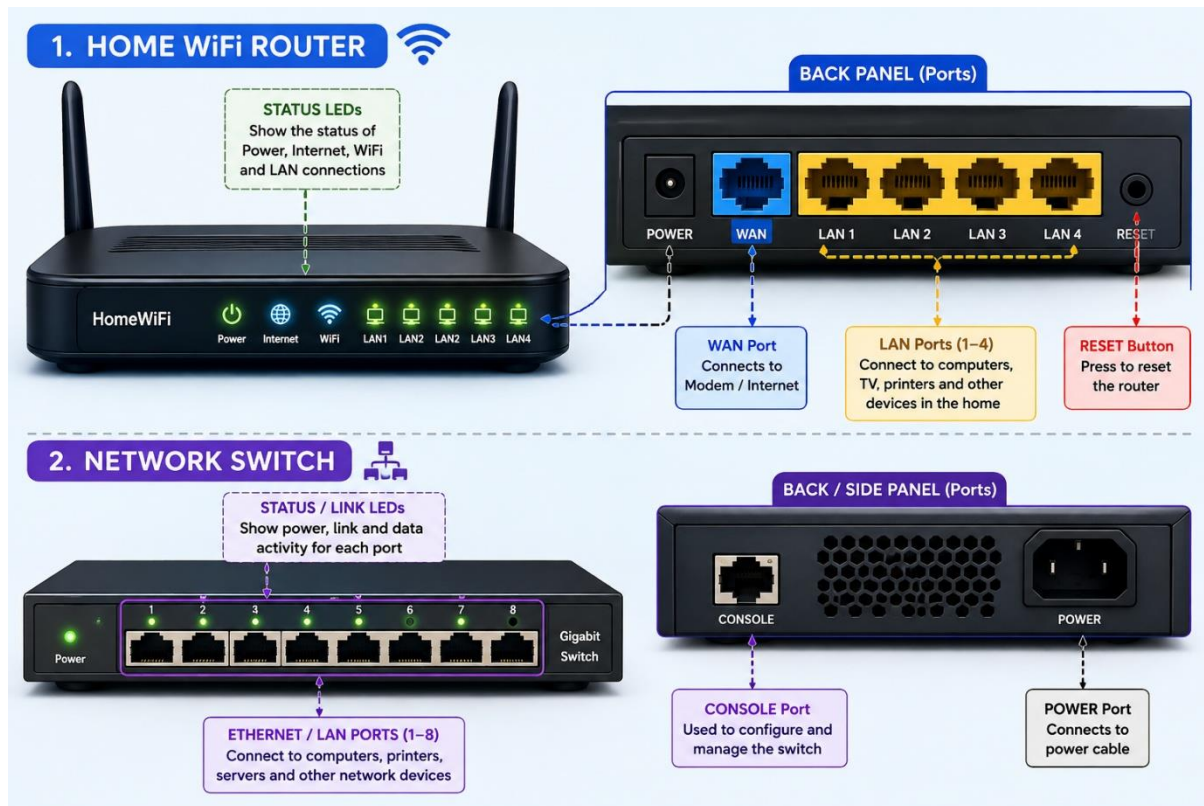
8/30/2026

Assignment - 2



Created By
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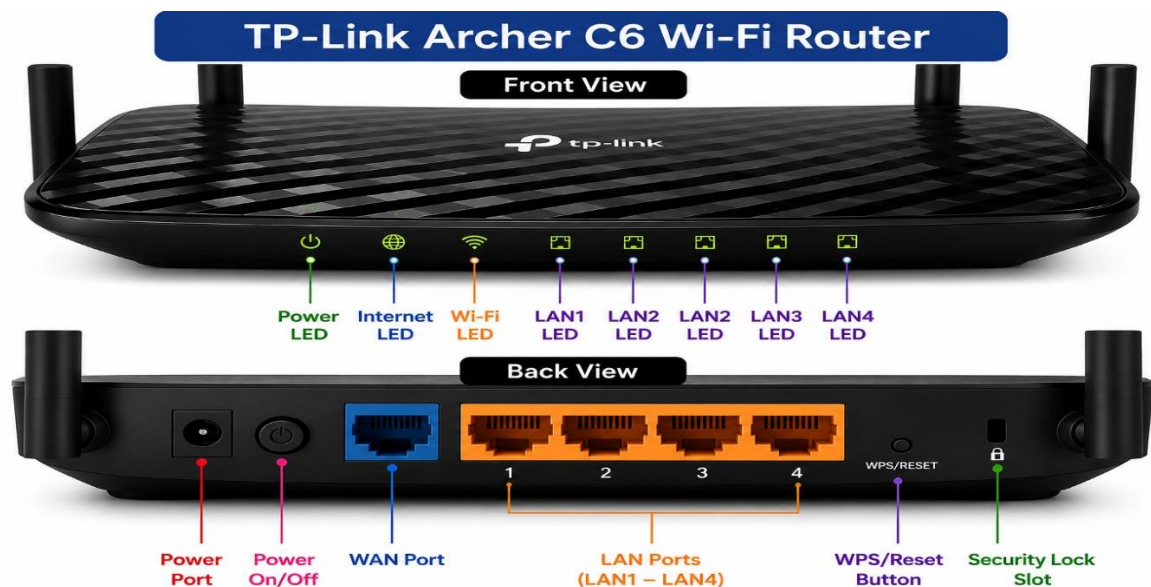
1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

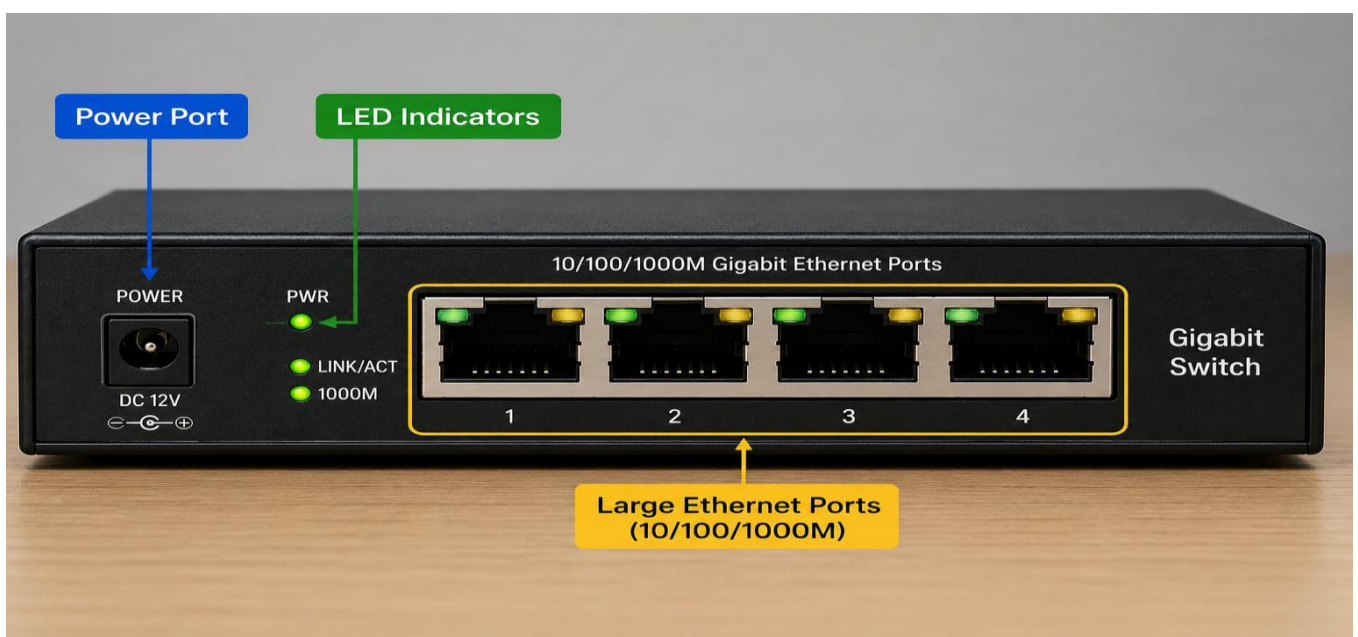
Difference	Router	Switch
1. Main Function	Connects different networks, such as a home network to the Internet.	Connects devices within the same network (LAN).
2. Address Used	Uses IP addresses to decide where network traffic should go.	Uses MAC addresses to forward data to the correct device.
3. Traffic Handling	Routes packets between different networks.	Forwards frames between devices on the same LAN.

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



Router – Short Explanation

- Power Port → Supplies power to the router.
- WAN Port → Connects the router to the modem/Internet.
- LAN Ports → Connect computers and other devices.
- LED Indicators → Show Power, Internet, Wi-Fi and LAN status.
- WPS/Reset Button → Used for quick Wi-Fi connection or resetting the router.



Network Switch – Short Explanation

- **Power Port:** Gives power to the switch.
- **LED Indicators:** Show power, connection, and data activity.
- **Ethernet Ports:** Connect computers and other network devices.
- **Port Numbers:** Identify each Ethernet port.
- **1000M LED:** Shows Gigabit (1000 Mbps) connection.
- **Main Work:** Connects multiple devices and forwards data to the correct device.

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Step 1: Connect ISP to Router

Connect the ISP Internet cable to the Router.

👉 The router manages the Internet connection, gives IP addresses, and controls traffic between the café network and the Internet.

Step 2: Connect Router to Switch

Connect the Router to the Switch using an Ethernet cable.

👉 The switch receives the network from the router.

Step 3: Connect 8 PCs to Switch

Connect all 8 gaming PCs to the Switch using Ethernet cables.

👉 The switch connects all PCs to the network.

Step 4: Start Gaming and Streaming

Now all 8 PCs can use the Internet for multiplayer games and IPL streaming.

Simple connection:

ISP → Router → Switch → 8 Gaming PCs